Putnam POTD November 24 2025

I. PROBLEM

Let D_n denote the value of the $(n-1)\times(n-1)$ determinant

$$\begin{vmatrix} 3 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 4 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 5 & 1 & \cdots & 1 \\ 1 & 1 & 1 & 6 & \cdots & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & 1 & \cdots & n+1 \end{vmatrix}.$$

Is the set $\{D_n/n!\}_{n\geq 2}$ bounded?

II. SOLUTION

First, subtract the **first row from the last row** to get:

$$D_n = \begin{vmatrix} 3 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 4 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 5 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & 1 & \cdots & n \\ -2 & 0 & 0 & 0 & \cdots & n \end{vmatrix}.$$

Expanding along the last row gives:

$$D_n = nD_{n-1} + (-2)(-1)^{(n-1)+1}A_{n-1} = nD_{n-1} + 2(-1)^n A_{n-1},$$

where A_{n-1} is the determinant of the $(n-2)\times(n-2)$ minor obtained by removing the first column and the last row. Explicitly:

$$A_{n-1} = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 4 & 1 & 1 & \cdots & 1 \\ 1 & 5 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots & \\ 1 & 1 & \cdots & n & 1 \end{vmatrix}.$$

Subtract the first row from every other row. The resulting matrix has non-zero entries on the diagonal shifted by one column (e.g., $3, 4, \ldots, n-1$) and zeros elsewhere below the first row, except for the last row which becomes $[0, \ldots, n-1, 0]$:

$$A_{n-1} = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 3 & 0 & 0 & \cdots & 0 \\ 0 & 4 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & \cdots & n-1 & 0 \end{vmatrix}.$$

Expanding along the last column, the only non-zero contribution comes from the top-right 1:

$$A_{n-1} = (-1)^{1+(n-2)} \cdot 1 \cdot \left(\prod_{k=3}^{n} (k-1) \right) = (-1)^{n-1} \frac{(n-1)!}{2}.$$

Substituting back into the expression for D_n:

$$D_n = nD_{n-1} + 2(-1)^n \left[(-1)^{n-1} \frac{(n-1)!}{2} \right].$$

Since $(-1)^n(-1)^{n-1} = -1$, this simplifies to:

$$D_n = nD_{n-1} + (n-1)!.$$

Let $B_n = D_n/n!$. Dividing the recurrence by $n!$:

$$\frac{D_n}{n!} = \frac{nD_{n-1}}{n!} + \frac{(n-1)!}{n!} \implies B_n = B_{n-1} + \frac{1}{n}.$$

For $n = 2$, $D_2 = |3| = 3$, so $B_2 = 3/2$. Note that B_n satisfies the recurrence of the Harmonic series H_n. Specifically, $B_n = H_n$. Since $\lim_{n\to\infty} H_n = \infty$, the set is unbounded.